

Machine Learning II Project Report

A. Introduction and Short Descriptive Analysis of Data

A stroke is a major medical emergency and a leading cause of disability worldwide. To save lives and prevent long-term damage, we need to catch high-risk patients early. Our goal was to build a tool that can predict if someone is likely to have a stroke based on their lifestyle, medical history, and age.

We used a real-world dataset from Kaggle and tested two main types of machine learning models: **Logistic Regression (Lasso/Ridge)** and **Support Vector Machines (SVM)**. A big challenge we faced was that fewer than 5% of people in the data actually had a stroke. Because of this "imbalance," we couldn't just look at how often the model was "right" (accuracy). We had to focus on **Sensitivity**, the model's ability to catch every single stroke victim without missing anyone.

Variables:

- gender(nominal): Gender of the patient
- age (continuous): Age in years
- hypertension (binary): Presence of hypertension (0 = No, 1 = Yes)
- heart_disease (binary): Presence of heart disease (0 = No, 1 = Yes)
- ever_married (nominal): Marital status
- work_type (nominal): Type of employment
- Residence_type (nominal): Type of residence (Urban/Rural)
- avg_glucose_level (continuous): Average blood glucose level
- bmi (continuous): Body Mass Index
- smoking_status (nominal): Smoking behavior
- stroke (binary target): Stroke occurrence (0 = No, 1 = Yes)

Stroke is the outcome variable, which is binary, making this a classification problem.

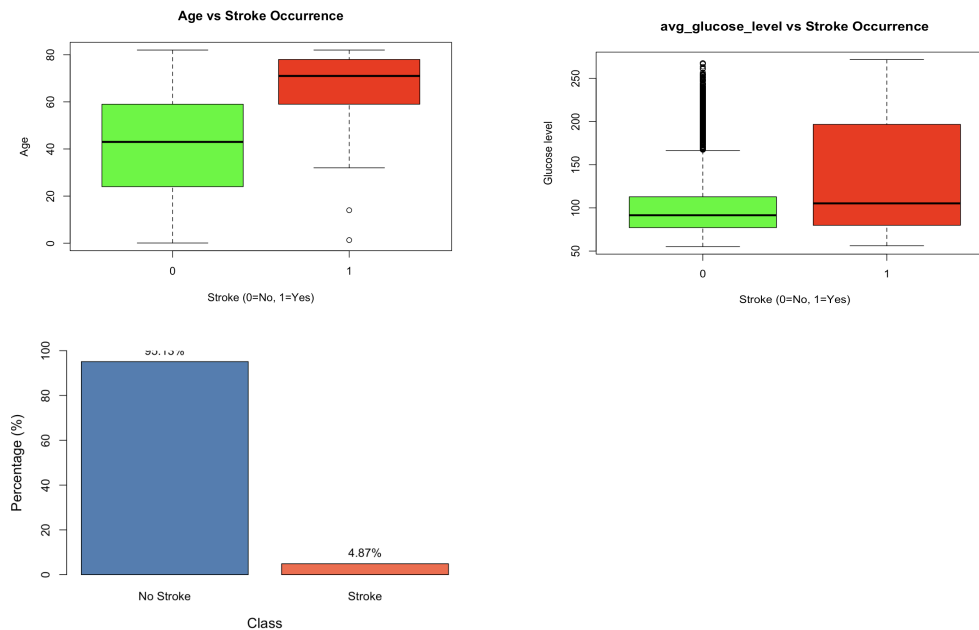
Data Description

We looked at 5,110 people, tracking details like age, smoking habits, job type, BMI, and blood sugar levels.

Observations:

- **Age:** The most obvious predictor. Stroke victims are significantly older (median age ~70) than healthy patients (median age ~43).
- **Glucose:** High blood sugar levels were a clear "red flag."
- **BMI:** Interestingly, BMI showed a lot of overlap between the two groups, suggesting it is a weaker predictor on its own.

- **Class Imbalance:** Our target variable was highly imbalanced (**less than 5% stroke and more than 95% no stroke**)



Data Pre-processing:

- The BMI feature was converted from a character to a numeric data type.
- Missing BMI values (201 instances) were addressed by imputation using the median.
- Categorical variables were transformed using one-hot encoding.

B. Brief Mathematical Overview of the Two ML Methods Used

Logistic Regression (ML1)

Logistic Regression with Regularization (Lasso & Ridge): We modeled the probability of a stroke using the logistic link function:

$$\ln\left(\frac{p}{1-p}\right) = \beta_0 + \sum \beta_i x_i$$

- **Ridge (L2):** Adds a penalty proportional to the *square* of the coefficients ($\lambda \sum \beta_j^2$). This shrinks all weights but keeps all variables in the model.
- **Lasso (L1):** Adds a penalty based on the *absolute* value of coefficients ($\lambda \sum |\beta_j|$). This is unique because it can shrink unimportant coefficients to exactly zero, effectively performing automatic feature selection.

Support Vector Machines (ML2)

SVM works by finding the "optimal hyperplane" that separates stroke patients from healthy ones with the widest possible margin. We focus on the **Support Vectors**, the data points closest to this boundary. The decision boundary is:

$$\mathbf{w}^\top \mathbf{x} + b = 0$$

Since our groups aren't perfectly separable, we used a **Soft Margin** with a cost parameter (C) to allow for some trade-off between margin width and classification errors. We also tested the **Radial Basis Function (RBF) kernel** to handle non-linear patterns by implicitly mapping data into a higher-dimensional space:

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$$

C. Description of Fitting Process and Model Comparison

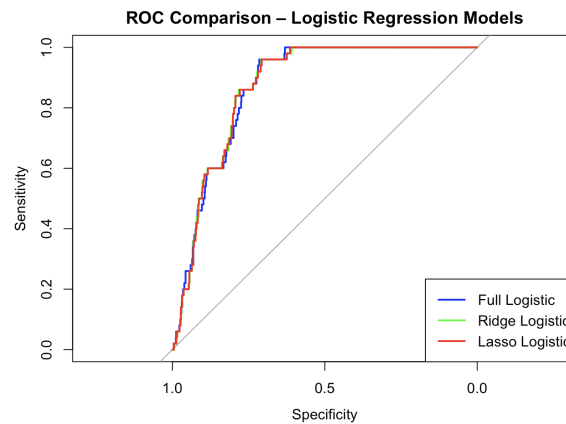
We split the data into a **60/20/20** ratio (Train/Validation/Test). We kept the final 20% of the data (test) away until the very end for a fair comparison between the models.

1. Logistic Regression (ML1)

We followed these steps to fit and optimize the **Logistic Regression** model:

- **Fitting:** We tried a few model variants:
 1. **Single-variable Model:** using only a single variable (age)
 2. **All-Feature Model:** this considers all variables in the dataset.
 3. **Ridge Regression Model:** regularized logistic regression using L2 regularization.
 4. **Lasso Regression Model:** regularized logistic regression using L1 regularization.
- **Hyperparameter Choice:** For Ridge and Lasso models, the **regularization strength (λ)** was automatically selected using 5-fold cross-validation to balance model fit and generalization.
- **Selection:** Ridge and Lasso Logistic Regression both improve stroke detection, achieving identical performance with 0.92 sensitivity and 0.87 AUC. Since both perform equally well, Lasso Logistic Regression is preferred because it produces a simpler and more interpretable model through feature selection.

##	Model	Accuracy	Sensitivity	Specificity	AUC
## 1	Full Logistic	0.733	0.90	0.724	0.869
## 2	Ridge Logistic	0.725	0.92	0.715	0.870
## 3	Lasso Logistic	0.725	0.92	0.715	0.870

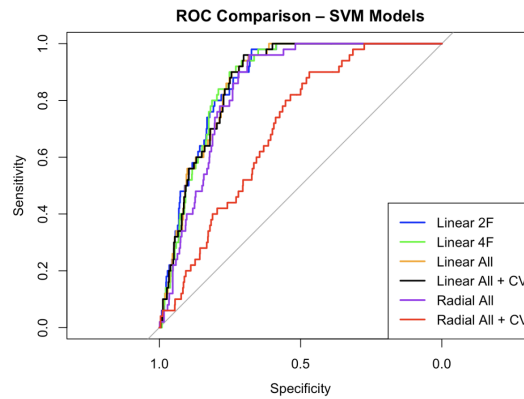


2. Support Vector Machines (ML2)

We followed these steps to fit and optimize the **Support Vector Machine** model:

- **Fitting:** We adopted a progressive approach to testing feature sets:
 1. **2-Feature Model:** Baseline using only Age and Glucose.
 2. **4-Feature Model:** Added BMI and Hypertension.
 3. **All-Feature Model:** Used Linear and Radial kernel considering all features.
 4. **Final Optimized Model:** Used Linear and Radial kernel considering all features with 5-fold cross-validation.
- **Hyperparameter Choice:** We tuned the **Cost (C)** parameter using a grid of values: 0.001, 0.01, 0.1, 0.5, 1, 5, 10, 50. We tested for both Linear and Radial kernels. The best model was a **Linear SVM** with a **Cost (C) = 10**.
- **Selection:** We found that while **Radial SVM** achieved high accuracy (84.1%), it is actually the most dangerous model to use because it misses nearly 76% of stroke cases. The **Linear SVM (All Features + CV)** was our winner, achieving a **Sensitivity of 0.94** and an **AUC of 0.868**.

##	Model	Accuracy	Sensitivity	Specificity	AUC
## 1	Linear SVM (2 features)	0.718	0.90	0.709	0.874
## 2	Linear SVM (4 features)	0.711	0.94	0.700	0.872
## 3	Linear SVM (All features)	0.708	0.94	0.697	0.869
## 4	Linear SVM (All + CV)	0.713	0.94	0.702	0.868
## 5	Radial SVM (All features)	0.728	0.88	0.720	0.845
## 6	Radial SVM (All + CV)	0.841	0.24	0.871	0.700



D. Assessment of Predicted Values

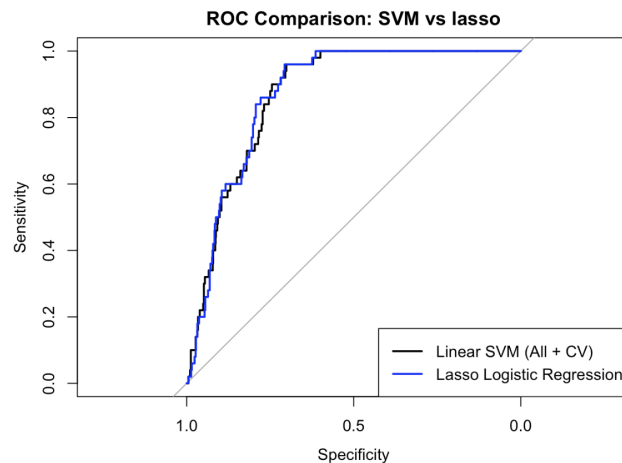
Our evaluation focused on two primary metrics that are critical for medical screening:

- **Sensitivity (Recall):** This tells us how good the tool is at finding *everyone* who actually had a stroke. In a hospital, a high score of **0.94** means we correctly catch 94 out of every 100 stroke patients. This is super important because missing a stroke can be deadly.
- **AUC (Area Under the Curve):** This is one simple score that shows the tool's overall power to tell the difference between "Stroke" and "No Stroke" patients. Our best tools scored around **0.87**, which means they are very good at separating the two groups.

E. Comparison of the Two Methods using Test Data

The final comparison of the two optimized models was conducted using the unseen test data.

Metric	Lasso Logistic	Linear SVM(with CV)
Accuracy	0.725	0.713
Sensitivity	0.920	0.940
AUC	0.870	0.868
Specificity	0.715	0.702



Conclusion:

- Both models show strong predictive performance for stroke prediction.
- **Linear SVM (All Features + CV)** performs best overall by achieving the **highest sensitivity (0.94)**, making it the most effective model for detecting stroke cases.
- In medical applications, minimizing false negatives is critical, and Linear SVM best satisfies this requirement.
- The model also demonstrates strong discriminative ability (**AUC = 0.868**), confirming reliable separation between stroke and non-stroke cases.
- Although Lasso Logistic Regression provides slightly higher accuracy, AUC(**0.870**) and specificity, **Linear SVM is preferred due to its superior sensitivity and suitability for medical screening.**

F. Interpretation of Trained Models using XAI Techniques

We first interpret each model individually using model-agnostic XAI techniques and then compare their explanations.

XAI Interpretation – Linear SVM (All + CV)

- Feature importance (PFI) identifies **age** as the strongest predictor, followed by **average glucose level**, work type, and smoking status.
- PDP and ALE plots show that stroke risk rises sharply after **55–60 years**, indicating a strong non-linear age effect.
- SHAP analysis shows that **older age and high glucose level** are the main factors pushing the prediction toward stroke for the selected individual.

XAI Interpretation – Lasso Logistic Regression

- Feature importance highlights **age and glucose level** as dominant predictors, with fewer variables retained due to Lasso's feature selection.
- PDP and ALE plots show a smooth, monotonic increase in stroke risk with age.

- SHAP values indicate that **age, glucose level, and heart disease** drive the individual prediction, while other features have minimal impact.

Overall Conclusion:

- Both models consistently identify **age** as the most important stroke risk factor, followed by **glucose level**.
- PDP and ALE plots reveal a clear age threshold around **55 years**, after which stroke risk increases rapidly.
- SHAP explanations show that **age and high glucose** dominate individual predictions in both models, confirming clinically meaningful and consistent decision logic.

G. Bibliography of Sources Used

Kaggle Stroke Prediction Dataset

<https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset>.

James et. al, "Introduction to Statistical Learning with R"

Moodle Slides, Exercise Solutions

R Package Docs: caret, pROC, dplyr, e1071, glmnet, iml, DALEX, MASS, DALEXtra, localModel, gridExtra